Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:20-12-24
Time: 3HRS GTM-2 Max. Marks: 300

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
C	D	A	B	A	B	A	B	D	B
11	12	13	14	15	16	17	18	19	20
B	D	C	C	C	A	C	D	D	C
21	22	23	24	25					
5	4	23	5	7					

CHEMISTRY

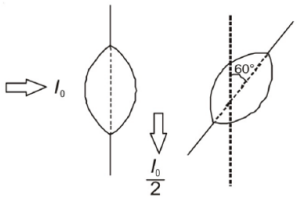
26	27	28	29	30	31	32	33	34	35
C	C	C	A	A	D	D	B	D	C
36	37	38	39	40	41	42	43	44	45
A	D	A	C	B	B	A	C	D	B
46	47	48	49	50					
1	4	16	1	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
D	B	D	A	A	B	B	C	C	B
61	62	63	64	65	66	67	68	69	70
A	A	A	A	A	C	D	A	C	D
71	72	73	74	75					
1	22	9	7	150					

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1.



$$I = \frac{I_0}{2}\cos^2 60° = \frac{I_0}{8}$$

$$\% \text{ transmitted light } \frac{\frac{I_0}{8}}{I_0} \times = \frac{100}{8} = 12.5\%$$

2. $$\Delta P = B\left|\frac{\Delta V}{V}\right|$$

$$2\times10^5 = B\frac{0.1}{100}$$

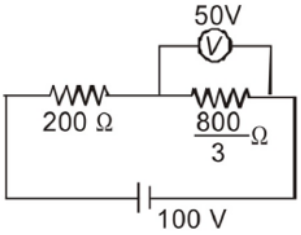
$$B = 2\times10^8\, Nm^{-2}$$

3. $$\vec{E} = \frac{200}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$$

$$\overrightarrow{r_{PQ}} = \vec{r_P} - \vec{r_Q} = \hat{i} - 2\hat{j}$$

$$V_P - V_Q = -\vec{E}.\overrightarrow{r_{PQ}} = \frac{200}{\sqrt{2}} volt$$

4. Voltage across $200\Omega = 50V$



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So, equivalent resistance across voltmeter is also 200Ω

i.e., $\dfrac{\dfrac{800}{3}.R}{\dfrac{800}{3}+R}=200$

$\Rightarrow R=800\Omega$

5. Breaking strength, F =breaking stress × area of cross-section

$$=\frac{S\times D^2}{4}$$

or $F\propto D^2$

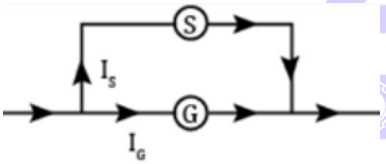
$$\therefore \frac{F_2}{F_1}=\left(\frac{D_2}{D_1}\right)^2$$

6. Average speed $=\dfrac{\text{Total Distance}}{\text{Total Time}}=\dfrac{20\pi}{6.28}$

$$=\frac{\text{Total Displacement}}{\text{Total Time}}=\frac{0}{6.28}=0\text{ m/s}$$

7. The correct option is A 18Ω

We know that a shunt resistance is connected in parallel to galvanometer as shown below. Let S and G be the resistance of shunt and galvanometer respectively.



From diagram

$$\frac{I_G}{I_S}=\frac{S}{G}$$

$$\Rightarrow G=\frac{I_S}{I_G}\times S$$

$$\Rightarrow G=\frac{50-20}{20}\times 12$$

[$\because I_G=20k$ and $I_S=50k-20k=30k$ where, k is proportionality constant]

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$\therefore G = 18\Omega$

8. $f_{static} = 2g\sin 37 = 12N$

9. $k = \frac{1}{2}mv^2 \Rightarrow \frac{dk}{dx} = \frac{1}{2}m\left(2v\frac{dv}{dk}\right) = ma$

$\Rightarrow a \propto$ slope of $k - x$ graph.

10. $W = P\left(l\sin 30°\right) = \frac{Pl}{2}$

11. $\frac{10}{\frac{12x}{12+x}} = \frac{50}{20}$

$12x == 48 + 4x$

$x = 6\Omega$

12. $V_{rms} = \sqrt{\frac{3RT}{2}}$

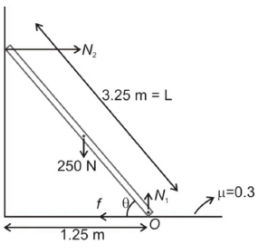
$(V_{rms})_H = \sqrt{\frac{3RT}{2}}$

$(V_{rms})_{He} = \sqrt{\frac{3RT}{4}}$

$\frac{(V_{rms})_H}{(V_{rms})_{He}} = \sqrt{2}$

$(V_{rms})_H = \sqrt{2}(V_{rms})_{He}$

13.



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$$\cos\theta = \frac{1.25}{3.25} = \frac{5}{13}$$

$$\sin\theta = \frac{12}{13}$$

For equilibrium; $N_1 = 240N$

$N_2 = f$

$\sum \tau_0 = 0$

$$\Rightarrow 240 \times \frac{L}{2}\cos\theta - N_2 \times L\sin\theta = 0$$

$$N_2 = \frac{240}{2}.\frac{\cos\theta}{\sin\theta} = \frac{120 \times 5}{12}N$$

Friction, $f = N_2 = 50N$

14. $W = 0$ and $Q = 0$

$\Rightarrow \Delta U = 0 \Rightarrow$ Isothermal process

$\Rightarrow P_1V_1 = P_2V_2$

$5 \times P = P_2 \times 11$

$$\Rightarrow P_2 = \frac{5P}{11}$$

15. $Q = \Delta U + W = \Delta U + \eta Q$

$\Rightarrow (1-\eta)Q = \Delta U$

$$(1-\eta)\frac{Q}{n\Delta T} = \frac{\Delta U}{n\Delta T}$$

$$(1-\eta).C = \frac{5}{2}R$$

$$\Rightarrow C = \frac{5R}{2(1-\eta)}$$

16. Conceptual

17. We know,

The least count for the standard screw gauge=0.01 mm

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The least count for the standard vernier caliper =0.1 mm

So, $L.C_{SG} < LC_{vc}$

18. Argument of trigonometric function is dimensionless

19. From conservation of momentum

$m_1 v_1 = m_2 v_2$

$m \propto A$

Hence, $A_1 v_1 = A_2 v_2$

$$\frac{A_1}{A_2} = \frac{v_2}{v_1} = \frac{1}{2}$$

$$R = R_0 A^{1/3}$$

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \frac{1}{2^{1/3}}$$

20. Einstein's photoelectric equation,

$$K_{\max} = \frac{hc}{\lambda} - \phi$$

$$\Rightarrow \quad K = \frac{hc}{\lambda_1} - \phi$$

$$\Rightarrow \quad 2K = \frac{hc}{\lambda_2} - \phi$$

$$\Rightarrow \quad 2\frac{hc}{\lambda_1} - 2\phi = \frac{hc}{\lambda_2} - \phi$$

$$\Rightarrow \quad \frac{2hc}{\lambda_1} = \frac{hc}{\lambda_2} + \phi \Rightarrow \frac{2}{\lambda_1} > \frac{1}{\lambda_2} \Rightarrow \lambda_1 < 2\lambda_2$$

21. At the bottom; $V = \sqrt{\frac{2gh}{1+\frac{2}{5}}} \Rightarrow V^2 = \frac{10gh}{7}$

For the smooth section

$$\frac{1}{2}mV^2 = mgh \Rightarrow \frac{1}{2} \times m \times \frac{10gh}{7} = mgh$$

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$$h_1 = \frac{5h}{7} = 5m$$

22. Case I : $\phi = \dfrac{q}{24\varepsilon_0}$

Case II: $k\phi = \dfrac{q}{6\varepsilon_0} \Rightarrow k = 4$

23. According to Bernoulli's principle, for points at same horizontal level

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2$$

$$\frac{1}{2}\rho V_B^2 = (P_A - P_B) + \frac{1}{2}\rho V_A^2 \quad \ldots(i)$$

But, $P_A - P_B = \rho gh$

$$\Rightarrow \frac{1}{2}\rho V_B^2 = \rho gh + \frac{1}{2}\rho V_A^2$$

$$\Rightarrow V_B^2 = 2gh + V_A^2$$

$$g = 10 ms^{-2}$$

$$V_A = 4cm\,s^{-1}$$

$$= 0.04\ ms^{-1}$$

$$h = 2.5\ mm = 2.5\times10^{-3} m$$

$$\Rightarrow V_B^2 = 2\times10\times2.5\times10^{-3} + (0.04)^2$$

$$= 51.6\times10^{-3} = 516\times10^{-4} m^2/s^2$$

$$\Rightarrow V_B = 22.72\times10^{-2}\ m/s$$

24. $y = a\sin\omega t\cos Kx$

$$y = \frac{1}{2}(2a\sin\omega t\cos Kx)$$

$\therefore$ Amplitude of component wave is $\dfrac{a}{2}$

25. $\tau = MB\sin 90°$

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$$= i \times \frac{1}{2}\left[\pi(2R)^2 + \pi R^2\right]B$$

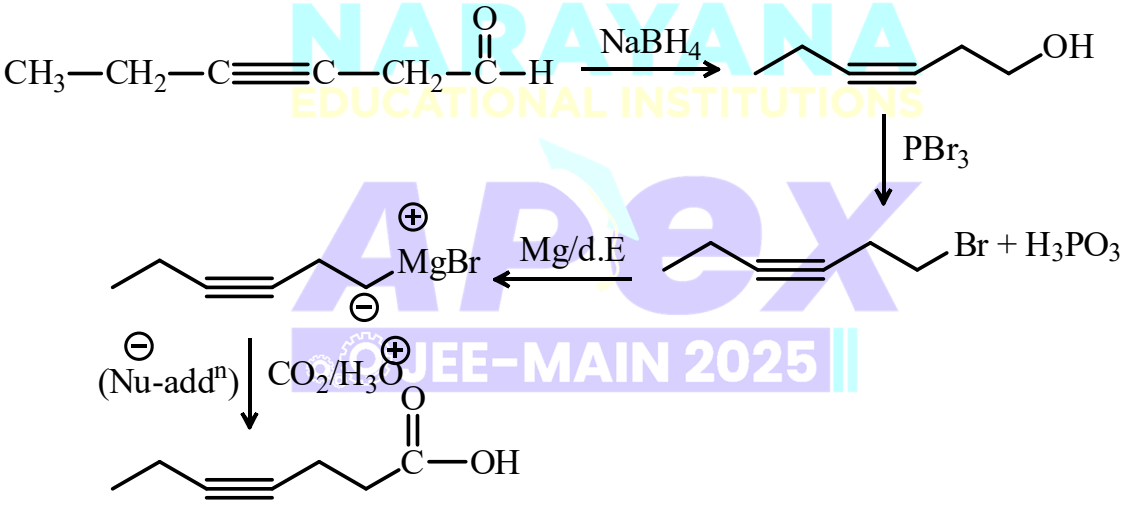
$$= \frac{5}{2}\pi R^2 iB$$

CHEMISTRY

26. Conceptual

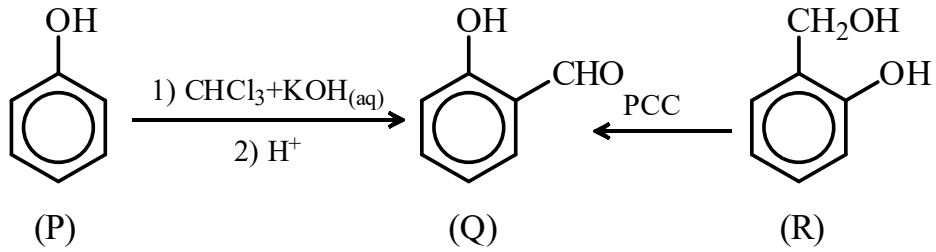
27.

28.



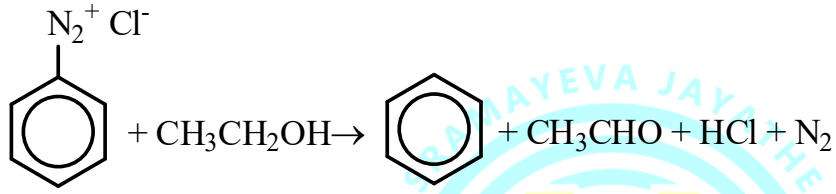
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(P) (Q) (R)

30. Refer NCERT page no – 413,414,415 12th class part II

31. 

32. Carboxylate ion is more stable than phenoxide ion

33. $\left[Ni(CO)_4\right] \rightarrow sp^3; \left[Ni(CN)_4\right]^{-2} \rightarrow dsp^2$

$\left[CO(CN)_6\right]^{-3} \rightarrow d^2sp^3; \left[COF_6\right]^{-3} \rightarrow sp^3d^2$

34. Compound 'A' exhibits plane of symmetry but compound 'B' do not exhibits any symmetry. Hence $A \rightarrow$ optically inactive $B \rightarrow$optically active

35. Refer $f-$block

36. (A)

Ion/molecule	No. of $\bar{e}$ in BMO	No of $\bar{e}$ in ABMO	Bond order
O_2^+	10	5	2.5
O_2	10	6	2
O_2^-	10	7	1.5
O_2^{2-}	10	8	1

37. Conceptual

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38.

Total energy needed to convert one Mg atom into Mg^{2+} (gas)

Ion = $IE_1 + IE_{II} = 22.681\ eV = 2188.6\ kJ\ mol^{-1}$

12 mg of $Mg = 0.5 \times 10^{-3}$ mole

39. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

x	$\bar{x}$	$\bar{x}$	
$5-x$	x	x	

Total no. of moles $= 5 + x + 2$ moles of N_2

$= 7 + x$

$\Rightarrow 7 + x = 10 \Rightarrow x = 3$

$PV = nRT \Rightarrow N = \dfrac{PV}{RT}$

$\Rightarrow n = \dfrac{2.46 \times 200}{0.082 \times 600} = \dfrac{2.46}{0.246}$

Total no. of moles =10moles

$n_{PCl_3} = 2 moles \Rightarrow C_{PCl_5} = \dfrac{2}{200}$

$K_P = K_C.(RT)^{\Delta n}$

$\Rightarrow K_P = \dfrac{9}{400} \times 0.082 \times 600$

$n_{PCl_3} = n_{Cl_2} = 3 moles \Rightarrow [PCl_3] = [Cl_2] = \dfrac{3}{200}$

$= \dfrac{54}{4} \times 0.082$

$\therefore = \dfrac{[PCl_3][Cl_2]}{[PCl_5]} = \dfrac{\frac{3}{200} \times \frac{3}{200}}{\frac{2}{200}} = \dfrac{9}{400}$

$= 1.1$ atm

40. $\Delta T_f = 0.45°C$ w_2 (acetic acid) $= 0.2\,g, w_1$(benzene) $= 20g$, $K_f = 5.12\,K\ kg\ mol^{-1}$

$\Delta T_f = i \times K_f \times m$

$\therefore \quad i = \dfrac{\Delta T_f}{K_f \times m} = \dfrac{0.45 \times 20 \times 60}{5.12 \times 0.2 \times 1000}$

$i = 0.527$

Initially: 1 mol 0

After time t: $(1-\alpha) mol \dfrac{\alpha}{2} \Rightarrow i = 1 - \alpha + \dfrac{\alpha}{2}$

$i = 1 - \dfrac{\alpha}{2}$ …(i)

On putting the value of i in equation (i), we get

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$$0.527 = 1 - \frac{\alpha}{2} \Rightarrow -0.946 = -\alpha$$

$$\alpha = 0.946$$

$\therefore$ Percentage association of acetic acid in benzene = 94.6%

41. Conceptual

42. Conceptual

43. Conceptual

44. Conceptual

45. Conceptual

46. (1)

47. Total =4 structure possible.
V-L-G-A
V-G-L-A
L-G-V-A
L-V-G-A

48. (16)

$$67\% = \frac{2.303}{0.693} \times t_{\frac{1}{2}} \log\left(\frac{100}{100-67}\right) = \frac{10}{3} \times t_{\frac{1}{2}} \times \log 3$$

$$= \frac{10}{3} \times t_{\frac{1}{2}} \times 0.48 = 1.6 \times t_{\frac{1}{2}} \Rightarrow x \times 10^{-1} = 16 \times 10^{-1} = 1.6 = 16$$

49. (1)

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{+2} + 10CO_2 + 3H_2O$$

$$\Rightarrow (5+16) - (2+10+8) = 1$$

50. $\frac{r_1}{r_2} = \frac{1}{8} = \frac{[M]^n}{[2M]^n} \Rightarrow n = 3$

MATHEMATICS

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51. $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^+} \dfrac{\tan^2 x}{x^2} = 1$

$\lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^+} \sqrt{\{x\}\cot\{x\}} = \sqrt{\cot 1}$

$\Rightarrow \cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2 = \cot^{-1}\cot 1 = 1$

$\tan^{-1}\left(\lim\limits_{x\to 0^+} f(x)\right) = \tan^{-1}(1) = \dfrac{\pi}{4}$

52. Let $z = x + iy$

Then, $\left|\dfrac{z-i}{z+2i}\right| = 1 \Rightarrow x^2(y-1)^2 = x^2 + (y+2)^2$

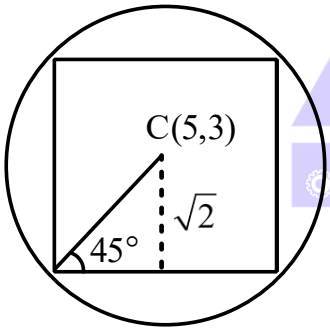
$\Rightarrow -2y + 1 = 4y + 4 \Rightarrow 6y = -3 \Rightarrow y = -\dfrac{1}{2}$

$\because |z| = \dfrac{5}{2} \Rightarrow x^2 + y^2 = \dfrac{25}{4} \Rightarrow x^2 = \dfrac{24}{4} = 6$

$\therefore z = x + iy \Rightarrow z = \pm\sqrt{6} - \dfrac{i}{2}$

$|z + 3i| = \sqrt{6 + \dfrac{25}{4}} = \sqrt{\dfrac{49}{4}} \Rightarrow |z + 3i| = \dfrac{7}{2}$

53.



One side of square is $y = x + k$

Distance of $(5,3)$ to the line $y = x + k$ is $\dfrac{|3 - 5 - k|}{\sqrt{2}} = \sqrt{2} \Rightarrow |-2 - k| = 2$

$\Rightarrow k = 0$ or $k = -4$

So lines are $y = x$ and $y = x - 4$

Now, solving these lines with circle

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$y = x$ and $x^2 + y^2 - 10x - 6y + 30 = 0$

$\Rightarrow 2x^2 - 16x + 30 = 0 \Rightarrow x^2 - 8x + 15 = 0$

$\Rightarrow x = 3,5 \Rightarrow x = 3, y = 3$

$x = 5, y = 5$

$y = x - 4$ and $x^2 + y^2 - 10x - 6y + 30 = 0$

$\Rightarrow x = 5, y = 1$

$x = 7, y = 3$

Points $(3,3)(5,5),(5,1)(7,3)$

$\sum_{i=1}^{4} x_i^2 + y_i^2 = 9 + 9 + 25 + 25 + 25 + 1 + 49 + 9 = 152$

54. Since four points having position vectors $\vec{a},\vec{b},\vec{c},\vec{d}$ are co-planar

So $\sin\alpha + 2\sin 2\beta + 3\sin 3\gamma = 1$

Let $\vec{p} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{q} = \sin\alpha\hat{i} + \sin 2\beta\hat{j} + \sin 3\gamma\hat{k}$

$\vec{p}.\vec{q} \le \left|\vec{p}\right|.\left|\vec{q}\right|$

$\Rightarrow \sin\alpha + 2\sin\beta + 3\sin 3\gamma \le \sqrt{14}.\sqrt{\sin^2\alpha + \sin^2 2\beta + \sin^2 3\gamma}$

$\Rightarrow \sin^2\alpha + \sin^2 2\beta + \sin^2 3\gamma \ge \frac{1}{14}$

55. $f\left(\frac{5x-3y}{2}\right) = \frac{5f(x) - 3f(y)}{2}$

$\Rightarrow f\left(\frac{5x-3y}{5-3}\right) = \frac{5f(x) - 3f(y)}{2}$

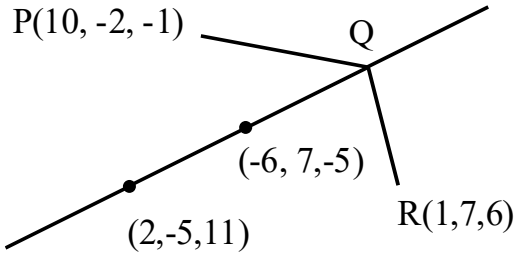
So $f(x)$ must be a linear function

$f(x) = ax + b \rightarrow f(x) = 2x + 3$

Period of $\sin(2x+3)$ is π

56.

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The given line : $\frac{x+6}{-8}=\frac{y-7}{12}=\frac{z+5}{-16}=\lambda$

Then co-ordinate of Q is,

$Q=(2\lambda-6,7-3\lambda,4\lambda-5)$

Now, $\overrightarrow{QR}=(2\lambda-7,-3\lambda,4\lambda-11)$

Since, QR is perpendicular to the line

$\therefore QR.(dr's \text{ of line})=0$

$\Rightarrow 4\lambda-14+9\lambda+16\lambda-44=0$

$\Rightarrow 29\lambda=58\Rightarrow\lambda=2$

$\therefore Q=(-2,1,3)$

$PQ=\sqrt{144+9+16}=\sqrt{169}=13$

57. Let n be a natural number.

$$f(x)=\begin{cases}\frac{-\sin(x-1)}{x-1} & x<-1\\ -(\sin 2+1) & x=-1\\ \cos 2\pi x & -1<x<1\\ 1 & x=1\\ \frac{-\sin(x-1)}{x-1} & x>1\end{cases}$$

$f(x)$ is discontinuous at $x=-1$ and $x=1$

58. Let matrix is $A=\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}$

Then,

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$$AA^T = \begin{bmatrix} a^2+b^2+c^2 & - & - \\ - & d^2+e^2+f^2 & - \\ - & - & g^2+h^2+i^2 \end{bmatrix}$$

Sum of diagonal entries,

$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2=9$

Case-I: Eight times zeroes and one time 3 = 9 cases

Case-II: Two times 2, one time 1 and six times zeroes

$=\frac{9!}{6!2!}=252$ cases

Case-III one time 2, five times 1 and three times zero

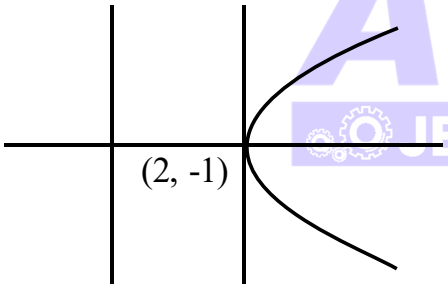
$=\frac{9!}{5!3!}=504$ cases

Case-IV Nine times ones = 1 case

$\therefore$ Total cases $= 9+252+504+1=766$

59. $\frac{1}{(101)!}\left[{}^{101}C_1+{}^{101}C_3+\ldots+{}^{101}C_{101}\right]=\frac{2^{100}}{(101)!}$

60.



We are given that directrix of the parabola is $4x-3y=21$

and vertex of the parabola is $(2,-1)$

Now $a=\frac{|8+3-21|}{5}=\frac{10}{5}=2$

$\therefore$ latus rectum of the parabola $=4a=8$

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61. $x\left(y^2.e^{xy}+e^{\frac{x}{y}}\right)dy=y\left(e^{\frac{x}{y}}-y^2.e^{xy}\right)dx$

$\Rightarrow y^2\left(xdy+ydx\right)e^{xy}=e^{\frac{x}{y}}\left(ydx-xdy\right)$

$\Rightarrow e^{xy}d\left(xy\right)=e^{x/y}\dfrac{ydx-xdy}{y^2}=e^{\frac{x}{y}}.d\left(\dfrac{x}{y}\right)$

$\Rightarrow e^{xy}=e^{\frac{x}{y}+c}$

62. You can have W, W, B or W, B, W or B, W, W

Required probability $=\frac{3}{4}.\frac{2}{4}.\frac{3}{4}+\frac{3}{4}.\frac{2}{4}.\frac{1}{4}+\frac{1}{4}.\frac{2}{4}.\frac{1}{4}=\frac{9}{32}+\frac{3}{32}+\frac{1}{32}=\frac{13}{32}$

63. $I=\int\dfrac{dx}{\left(x+4\right)^{8/7}\left(x-3\right)^{6/7}}\Rightarrow\int\dfrac{dx}{\left(x+4\right)^2\left(\dfrac{x-3}{x+4}\right)^{6/7}}$

Let $\dfrac{x-3}{x+4}=t^7$

Differentiate on both sides, we get

$\left(\dfrac{\left(x+4\right)-\left(x-3\right)}{\left(x+4\right)^2}\right)dx=7t^6dt\Rightarrow\dfrac{7}{\left(x+4\right)^2}dx=7t^6dt$

Hence, $I=\int t^{-6}t^6dt=t+C=\left(\dfrac{x-3}{x+4}\right)^{\frac{1}{7}}+C$

64. $\therefore x={}^5C_4\,4!=5.24=120$

$y={}^{20}C_4.4!=20\times19\times18\times17$

65. $y-$axis is angular bisector

The image $\left(2,1\right)$ w.r.t $y-$ axis is $\left(-2,1\right)$ lies on reflected ray is $2x-7y+11=0$

This is one of the directrix the equation of other directrix is $2x-7y+\lambda=0$

The equation of other directrix and nearest focus is

$\dfrac{a}{e}-ae=\dfrac{16}{\sqrt{53}}$ $\qquad$ $3a-\dfrac{a}{3}=\dfrac{16}{\sqrt{53}}$

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$$\frac{8a}{3}=\frac{16}{\sqrt{53}} \qquad a=\frac{16}{\sqrt{53}}$$

$$\left|\frac{\lambda-11}{\sqrt{53}}\right|=\frac{36}{\sqrt{53}}=\frac{2a}{e}$$

$$\lambda-11=\pm 36$$

66. $e^x F(x)=\int_3^x\left(3t^2+2t+4F^1t\right)dt$

Let $y=F(x)$

$$\frac{dy}{dx}+\frac{e^x}{e^x-4}y=\frac{3x^2+2x}{e^x-4}$$

$\text{I.F}=e^x-4$

Solution $y\left(e^x-4\right)=\int\left(3x^2+2x\right)dx$

$$y\left(e^x-4\right)=x^3+x^2+c$$

$$F(3)=0\Rightarrow y(3)=0$$

$$c=-36 \qquad F(x)=\frac{x^3+x^2-36}{e^x-4}$$

67. $u=\dfrac{2z+i}{z-ki}=\dfrac{2(x+iy)+i}{x+iy-ki}=\dfrac{2x+i(2y+1)}{x+i(y-k)}\times\dfrac{x-i(y-k)}{x-i(y-k)}$

$$\text{Re}\,u=\frac{2x^2+(2y+1)(y-k)}{x^2+(y-k)^2}$$

$$\text{Ing}\,u=\frac{x(2y+1)-2x(y-k)}{x^2+(y-k)^2}$$

$\text{Re}\,u+\text{Ing}\,u=1$ ....(1)

Put on $y-$ axis $\Rightarrow x=0$ put in (1) $\dfrac{(2y+1)(y-k)}{(y-k)^2}=1$

$$\Rightarrow(y-k)(y+1+k)=0\Rightarrow y=k,-(1+k)$$

$P(0,k),Q(0,-1,-k) \qquad PQ=5\Rightarrow|2K+1|=5$

$2K=4,-6\Rightarrow K=2,-3;\quad K>0\Rightarrow K=2.$

68. $\log_{\frac{1}{\sqrt{3}}}\left(\dfrac{x^2-x+1}{x+2}\right)>-2$

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$\Rightarrow 0 < \dfrac{x^2 - x + 1}{x + 2} < 3$

Since $\Rightarrow 0 < \dfrac{x^2 - x + 1}{x + 2} < 3$, hence domain is $x + 2 > 0$

Now $\dfrac{x^2 - x + 1}{x + 2} < 3$

$x^2 - x + 1 > 3x + 6$

$\Rightarrow \quad x^2 - 4x - 5 < 0$

$\Rightarrow \quad (x - 5)(x + 1) < 0$

Hence $x \in (-1, 5)$ also domain is $x > -2$

$\Rightarrow \quad -1 < x < 5$

69. $(\bar{x}) = \dfrac{\text{sum of quantities}}{n} = \dfrac{\frac{n}{2}(a + l)}{n}$

$= \dfrac{1}{2}[1 + 1 + 100d] = 1 + 50d$

$\therefore \text{MD} = \dfrac{1}{n}\sum |x_i - \bar{x}| \Rightarrow 255$

$= \dfrac{2d}{101}\left(\dfrac{50 \times 51}{2}\right)$

$\therefore d = \dfrac{255 \times 101}{51 \times 50} = 10.1$

70. $\vec{a}.\vec{b} = 0$

$\vec{c}.\vec{a} = \vec{c}.\vec{b} = \cos\theta$

$\vec{c}.\vec{a} = \lambda\left(\vec{a}.\vec{a} + \vec{a}.\vec{b}\right) + \mu\vec{a}.\left(\vec{a} \times \vec{b}\right)$

$\cos\theta = \lambda$

$c.c = \lambda^2 (a + b)^2 + \mu^2 (1.1.\sin 90°)^2 + 2\lambda\mu\left\{\left(\vec{a} + \vec{b}\right).\left(\vec{a} \times \vec{b}\right)\right\}$

$1 = \lambda^2 (1 + 1 + 2.0) + \mu^2 + 2\lambda\mu(0) \qquad \left[\because |\vec{a}| = 1, |\vec{b}| = 1 \text{ and } \vec{a}.\vec{b} = 0\right]$

$1 = 2\cos^2\theta + \mu^2$

$\mu^2 = 1 - 2\cos^2\theta \geq 0$

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$$-\frac{1}{\sqrt{2}} \le \cos\theta \le \frac{1}{\sqrt{2}} \Rightarrow \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

71. $$f''(x) = \begin{cases} a^2 x e^{ex} + 2ae^{ax} & ; \quad x \le 0 \\ 2a - 6x & ; \quad x > 0 \end{cases}$$

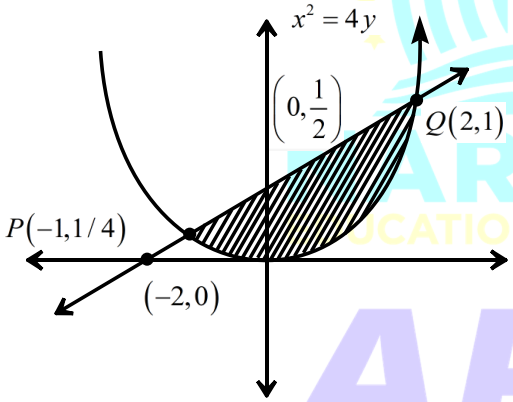
$f'(x)$ increasing if $(ax+2)ae^{ax} \ge 0$ and $2a - 6x \ge 0$

72. $$12 \times 2\left(\int_0^1 (x^2 - 3x + 2)dx - \int_1^{3/2} (x^2 - 3x + 2)dx\right)$$

$$= 24\left(\left(\frac{x^2}{3} - \frac{3}{2}x^2 + 2x\right)_0^1 - \left(\frac{x^3}{3} - \frac{3x^3}{2} + 2x\right)_1^{3/2}\right)$$

$$= 24\left(\frac{5}{6} + \frac{1}{12}\right) = 24 \times \frac{11}{12} = 22$$

73.



Let points of intersection of the curve and the line be P and Q

$$x^2 = 4\left(\frac{x+2}{4}\right) \Rightarrow x = 2, -1$$

Point are $(2,1)$ and $\left(-1, \frac{1}{4}\right)$

$$\text{Area} = \int_{-1}^{2}\left[\left(\frac{x+2}{4}\right) - \left(\frac{x^2}{4}\right)\right]dx = \left[\frac{x^2}{8} + \frac{1}{2}x - \frac{x^3}{12}\right]_{-1}^{3}$$

$$= \left(\frac{1}{2} + 1 - \frac{2}{3}\right) - \left(\frac{1}{8} - \frac{1}{2} + \frac{1}{12}\right) = \frac{9}{8}$$

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74. $D=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 2 & 5 & \lambda\end{vmatrix}=\lambda-8\Rightarrow\lambda=8 \quad D_3=\begin{vmatrix}1 & 1 & 6\\ 1 & 2 & 14\\ 2 & 5 & \mu\end{vmatrix}=\mu-36=0\Rightarrow\mu=36$

75. P (at least one hits the target)

$=1-P$ (none of them hits the target)

$=1-\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{8}\right)$

$=1-\frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}\times\frac{7}{8}=1-\frac{7}{32}=\frac{25}{32}$

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